3. ISLR 4.8.3

Question: We have K classes, each with X such that $X \sim N(\mu_k, \sigma_k^2)$. Own mean μ_k & variance σ_k^2.
Prove that here Bayes classifier is not linear.

Solution:

Using Bayes classifier

Assign $X = x$ to class k that maximizes posterior probablity.

$$\hat{y} = \arg\max_k P(Y = k \mid X = x)$$

$$P(Y = k \mid X = x) \propto P(X = x \mid Y = k) \cdot P(Y = k)$$

By Baye's Theorem we get

$$P(Y = k \mid X = x) = \frac{f_k(x)\,\pi_k}{\sum_{j=1}^{k} f_j(x) \cdot \pi_j}$$

Here $f_k(x)$ is class condition density of class K.
π_k is the prior probability of class K.

$\therefore$ we get

$$\hat{y} = \arg\max_k f_k(x) \cdot \pi_k$$

Using Normal Distribution we get :

$$f_K(x) = \frac{1}{\sqrt{2\pi\sigma_K^2}} \cdot \exp\left[-\frac{(x-\mu_K)^2}{2\sigma_K^2}\right]$$

(Likelihood for class K.)

$$f_K(x)\,\pi_K = \frac{\pi_K}{\sqrt{2\pi\sigma_K^2}} \exp\left[-\frac{(x-\mu_K)^2}{2\sigma_K^2}\right]$$

Applying log on both the sides.

$$\log_e(f_K(x)\,\pi_K) = \log \pi_K - \frac{1}{2}\cdot\log(2\pi\sigma_K^2) - \frac{(x-\mu_K)^2}{2\sigma_K^2}$$

Decision Boundary

Between classes K & j, set $\delta_K(x) = \delta_j(x)$.

$$\frac{(x-\mu_j)^2}{2\sigma_j^2} - \frac{(x-\mu_K)^2}{2\sigma_K^2} = \text{constant}$$

The equation is thus quadratic equation in x cause of x^2 terms.

Therefore if $\sigma_K^2 \neq \sigma_j^2$, decision boundary is quadratic bayes classifier is not linear unless all σ_K^2 are equal.

Hence proved it's quadratic equation in terms of x.